Segmentation of financial clients by attitudes and behavior: a comparison between Switzerland and Vietnam

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Abstract

Purpose – This paper aims to investigate segmentations by finance-related attitudes and behavior of financial consumers in Switzerland and Vietnam.

Design/methodology/approach – We replicated the questionnaire measuring attitudes toward financial affairs as used in the study of Fünfgeld and Wang (2009). In order to extract factors and identify profiles of financial customers, we applied the same techniques including factor analysis and a two-step cluster analysis (Ward and K-means) to both samples. Linear regressions were used to examine the impact of socio-demographic variables.

Findings – Factor analysis reveals surprisingly similar underlying dimensions of financial attitudes and behavior for participants in both countries. The five clusters based on these dimensions exhibit strikingly similar characteristics. We also find that socio-demographic variables such as gender, age, and education significantly vary across groups.

Research limitations/implications – There are small differences in the composition of the revealed dimensions between the two countries, mainly a mixture of intuitive decisions and free-spending aspects. Future research may investigate more countries with comparable samples to further test the reliabilities and validities of the scales.

Practical implications – The findings suggest that the instruments as developed by Fünfgeld and Wang (2009) can be used not only in highly-developed Western markets, but also in non-Western and emerging markets. Such instruments can be very useful for offering financial advice based on market segmentation in a global market.

Originality/values – The study provides empirical evidence on universal segmentation variables that reflect financial consumers' psychological traits and attitudes toward saving and spending. Furthermore, the study stresses the need for a broad use of those variables when studying consumer decision-making in international financial markets.

Keywords Universal segmentation variables, cross-national clusters, personal finance, financial advice.

1 Introduction

Applying existing marketing strategies to emerging markets is not always straightforward: Sheth (2011) argues that there are large differences in marketing conditions between developed and emerging markets at both theoretical and practical levels. Accordingly, if marketers want their marketing strategies to remain relevant for emerging markets, they have to reconsider core assumptions of marketing and then develop new theories, models, and methods. Otherwise, consumers in emerging markets will not respond efficiently to the marketing strategies transferred from developed countries. According to IMF (2016a), emerging markets have been integrating at an increasing rate into the global economy recently. Therefore, a question arises: How big is the gap between emerging and developed markets regarding the application of marketing strategies?

In this paper, we focus on one particular case, market segmentation strategies in the financial sector and analyze whether there are similarities in the segmentation of financial consumers between affluent and emerging markets and if so, which segmentation variables should be used for categorizing those markets. These findings are important for multinational institutions that offer financial advice to customers and want to adapt their advice to different types of clients. The findings also contribute to the debate about which kind of market segmentation is effective for classifying retail financial clients irrespective of their identities, economic and cultural roots.

In the current study, we examine the external validity of the underlying dimensions of individual day-to-day financial behavior as proposed by Fünfgeld and Wang (2009). Their original study was conducted in Switzerland, a highly developed market with European culture. We extend this research to an entirely different country, Vietnam, an emerging market with East-Asian culture. The replication with an extension (Toncar and Munch 2010) allows us to study whether segmentation parameters still keep their validity in a different cultural and economic context, and if so, whether different

segments exhibit similar demographic features.

We choose to replicate the study of Fünfgeld and Wang (2009) because their five measurement scales cover many aspects of financial consumer behavior and are strongly related to some attempts to investigate personal financial management. In particular, the first component which includes four statements is quite similar to the "Anxiety" construct of Money Attitudes Scales as developed by Lim and Teo (1997). The second factor (Interests in financial matters) comes close to Loix et al. (2005) who introduce a measurement scale and construct of orientation towards personal financial management with two components: financial information and personal financial planning. The third factor (Intuitive decisions) can be found in the study of Scott and Bruce (1995) who develop a consistent and sound instrument of decision-making styles. The last two factors are related to fundamental dimensions of individual financial behavior, including saving and spending tendency.

Moreover, the scales were used as segmentation variables for identifying five distinct subgroups of Swiss participants with different levels of financial competence and different needs for financial advice and products. The predictive validity of the scales is further supported in their follow-up study: Fünfgeld and Wang (2017) in which they examine whether individual differences in daily financial behavior can predict risk and time preferences while conducting laboratory experiments for 161 Swiss participants with diverse backgrounds. More specifically, they find that risk aversion is positively associated with *needs for precautionary saving* and is inversely correlated with *free spending* and *an interest in financial matters*. More consistent time discounting is linked to *an interest in finance* and *needs for precautionary saving* whereas *anxiety concerning financial issues* predicts hyperbolic discounting behavior.

Our results show that most of the five underlying dimensions of Swiss individual financial attitudes can be found in our Vietnamese sample. Further, the clusters based on these instruments exhibit strikingly similar characteristics. We also find that socio-demographic variables, such as gender, age, and education, significantly vary across groups.

Our study provides several contributions. Firstly, this is the first study which is using individual financial attitudes and behavior as market segmentation variables across nations. Secondly, the findings of this study provide insights into the merits of universal segmentation bases, thereby offering an assessment of their relevance and application to many other countries. Finally, the findings of this study on cluster characteristics will allow marketing managers to adapt their financial advice to specific financial needs of investors from different markets and cultures.

This study is organized as follows: Section 2 provides a theoretical background of market segmentation in domestic and international financial markets. Section 3 shows the data sample and choice of Vietnam as a replication study for the assessment of market segmentation. Section 4 analyses empirical findings and robustness tests. Section 5 provides the interpretation of primary results and discusses the limitation of the study.

2 Theoretical background

2.1 *Market segmentation in the financial market*

Market segmentation, initially proposed by Smith (1956), has been one of the most common marketing strategies in managing diverse customer needs and targeting marketing resources (LaPlaca 1997; McDonald and Dunbar 2004; Weinstein 2004). The underlying principle of this strategy is that heterogeneities in a market can be managed by grouping customers with similar characteristics into homogeneous groups. The information of the homogeneous segments, in turn, is useful for market makers to match their products and capabilities with consumer requirements (Baker and Hart 2008; Piercy et al. 2012) and develop customized marketing strategies for selected segments (Anderson and Vincze 2000). In a business sense, market segmentation is attractive because it can assist in building customer retention and loyalty (Weinstein 2002) and increasing customer satisfaction levels (Harrison and Ansell 2002), both of which positively affect firm performance (Martenson 2008; Williams and Naumann 2011).

In the finance industry, market segmentation has attracted greater attention since an increase in the competitive climate of the financial sector took place after the deregulation of the 1980s. The relaxation of legal restriction forced many financial institutions to redefine their marketplaces and develop positive strategies in an attempt to maintain competitive edges and profit objectives (Harrison 1994). Changes in the market environment also increased the importance of identifying profitable markets and requirements for developing customer orientation and gaining more intimate knowledge of markets (Gunnarsson and Wahlund 1997). Thus, segmentation schemes, which "recognize the importance of concentrating on the needs of homogeneous groups within a larger heterogeneous market," have become preferences for most providers of financial services (Joseph and Yorke 1991).

In applying market segmentation to the finance industry, market makers can use customer socio-demographic, psychographic and behavioral variables as the segmentation basis for classifying customers (Kotler 2001). Yet,

financial marketers have still heavily relied on socio-demographic features (a priori approach), which are easy to observe and implement in the market (Meadows and Dibb 1998; Alfansi and Sargeant 2000; Machauer and Morgner 2001). However, the prediction of customers needs can not be assumed by using the "priori approach" because of its single dimension (Machauer and Morgner 2001), the lock of guarantee of homogeneous behavior and lifestyles among customers in the same demographic category (Meadows and Dibb 1998; Kotler 2009) and the lock of evidence of differences in purchase behavior, brand preference, product purchasing, innovation adoption, channel use and technology uptake among demographic groups (Piercy et al. 2012). Therefore, to address the limitations of the prior approach, researchers have increasingly suggested multiple bases or a hybrid approach as solutions for market segmentation in financial markets (Machauer and Morgner 2001; Soper 2002). Under the hybrid approach, homogeneous groups can be generated by multiple dimensions of segmentation (Walsh et al. 2001). As different types of segment bases have their inherent strengths, the hybrid approach, by integrating key features, advantages, and benefits of each dimension, can improve the potential of identifying homogeneous groups. This can also help "offering each customer the right product, for the right reasons, through the right channel and at the right time" (Soper 2002).

Previous studies in the finance industry have documented various combinations of segment variables for the hybrid scheme, but most of them have integrated socio-demographic with other behavioral variables: McDougall and Levesqu (1994) classify Canadian customers based on service quality, the usage of financial products and demographic variables. Alfansi and Sargeant (2000) link personal characteristics to desired benefits of using banking services as the primary basis for classifying private customers in Indonesia. Due to its unrepresentative sample, the study captures only one demographic variable (occupation) which relates to cluster membership. However, the study of Fünfgeld and Wang (2009) has ascertained the rationale of the hybrid approach when extending the sample to the general population of Switzerland and taking financial attitudes and behavior into account. The study succeeds in not only generating sophisticated dimensions of individual finance-related attitudes but also reveals marketing implications for five distinct subgroups of financial clients based on those scales. Followed by that study, some researchers have recently utilized segment variables for the hybrid approach. For example, Liang and Pei-Ching (2015) categorize customers who use internet banking by their perceptions, behavior, loyalty, satisfaction and demographic characteristics whereas Makgosa et al. (2016) base this approach on individual saving attitudes, behavior, psychographic and demographic factors.

We will apply the hybrid approach in this study and examine the integrated inventory of individual financial attitudes, behavior, and socio-demographic bases, proposed by Fünfgeld and Wang (2009) in the different context of an emerging market. By replicating the scales, this study will give guidance for marketers in applying them to many other countries. Whether the hybrid approach can be applied to the international market setting will be highlighted in the following sections.

2.2 *International Market Segmentation*

In an international setting, marketers have developed distinct approaches to structure the heterogeneity that exists among customers and nations by identifying similar segments of customers across countries. Traditionally, they draw cross-national segments based on observable data at national level according to similarities and dissimilarities in economic development, cultural values, and other shared macro traits (see Budeva and Mullen (2014) for a recent overview). Under this approach, each country is treated as a market unit, and after clustering, marketing strategies will be developed to appeal and satisfy customers within segments of countries. Thus, this approach can help marketers to benefit from economies of scale after offering standardized marketing strategies to large countries and also can highly succeed in such plans when consumers share similar needs within the targeted homogeneous segment (Yip 2001; Steenkamp and Ter Hofstede 2002). However, marketing scholars raise concern regarding the relevance of this approach when assuming that countries are homogeneous units (Kale and Sudharshan 1987; Hassan and Katsanis 1991; Jain 2001) and segmenting markets without considering behavioral bases (Helsen et al. 1993; Nachum 1994).

In contrast, the contemporary approach of grouping customers across countries according to their characteristics (such as demographics, lifestyles, psychographics, values, and attitudes) has a greater prospect of success in profitably satisfying customers' needs and demands (Cleveland et al. 2011). By using data at individual level, this approach is likely to provide relevant information for managerial decisions (Burgess 1992) and contribute to the need of using customer orientation in any marketing segmentation method (Hassan et al. 2003). Samli (2012) has stressed that "International markets must be segmented by consumer needs or behavior." Additionally, the global convergence of customer needs and demands as impacted by the growth of globalisation and integration of the marketplace, provides support for this approach with a high likelihood of targeting at global niches (Solberg 2002; Nelson and Paek 2007). Studies regarding cross-national segments of consumers, however, are under-represented, presumably because of the low avail-

ability and high cost of consumer databases (Cleveland and Laroche 2007; Merz et al. 2008), together with the high risk of measurement, sampling bias and construct nonequivalence (Steenkamp and Ter Hofstede 2002).

Some researchers have suggested a promising approach with considering country-level data as the preliminary step in the two-stage process of global marketing segmentation (Steenkamp and Ter Hofstede 2002; Cavusgil et al. 2004; Hassan and Craft 2012). Because there are vast differences between countries regarding socio-economic, cultural and other dimensions, this step can offer comparative information about countries by eliminating less attractive markets and simultaneously determining attractive entry ones. In the next step, data at the individual level can be collected for grouping customers within selected geographic clusters into homogeneous segments. Apparently, this two-stage process can use different types of segmentation bases at both country and individual levels, which is likely to profit from their respective strengths (Steenkamp and Ter Hofstede 2002; Hassan et al. 2003). Also, this strategic approach can better form segments that share similarities in preferences, needs, and consumption patterns than the traditional approach (Steenkamp et al. 1999) and thereby can contribute to the success of a global marketing strategy (Hassan and Craft 2005). Notwithstanding potential outcomes, the practicability and robustness of this approach need to be tested on the two standard criteria of segmentation validity: accessibility and substantiality (Hassan and Craft 2012).

In the finance industry, there are very few studies concerning individual financial needs both in the single and cross-country level. To our knowledge, the study of Fünfgeld and Wang (2009) for Swiss financial customers is among the first that has studied financial consumers on a broader range of their financial attitudes and behavior since it covered not only individual attitudes toward financial matters (anxiety, interest in financial matters, and intuitive decisions), but also precautionary saving and impulsive spending. Our study is the first to test the external validity of their instruments in the setting of cross-country comparison.

3 Methodology

We choose to replicate segmentations of individual investors in Vietnam because Vietnam considerably differs from Switzerland in terms of socio-economic conditions and cultural values. Firstly, Vietnam is situated in the heart of Asia, governed by a Communist party and ranked the 15th most populous country in the world in 2017 (WorldBank 2017) whereas Switzerland is in Europe, has a direct democracy and a small population size. Secondly,

Vietnam has one of the fastest growing economies in the world with an average GDP growth of 6.87% between 1990 and 2015 (IMF 2016b) and is a middle-income country since 2009. On the contrary, Switzerland is one of the world's most prosperous countries in the recorded history with an average GDP per capita of 58,647 USD over the years 1990-2015 (IMF 2016b). Lastly, Vietnam and Switzerland also differ in their cultures. This can be seen, e.g., by comparing Hofstede's cultural dimensions: Switzerland, especially the German speaking part where Fünfgeld and Wang (2009) was conducted, represents a culture of individualism, masculinity, low power distance and high uncertainty avoidance, along with indulgence whereas Vietnam exhibits that of collectivism, femininity, high power distance, and low uncertainty avoidance, together with restraint (Hofstede et al. 2010). Also in the measurements of the World Value Survey, both countries differ substantially (WVS 2017). Thus, the market segmentation of financial consumers based on Swiss attitudes and behavior, if found in Vietnam, can provide a high assessment of its relevance and application to many other countries and also reflect the emergence of a homogeneous global consumer culture as proposed by Cleveland and Laroche (2007). Furthermore, the comparison of segments of financial clients between Vietnam and Switzerland can provide insights into contextual causes for variations in individual needs and demands between emerging and advanced markets, which seems to be essential for marketing research (Sheth 2011).

Questionnaire design

To assess whether there are similarities in the segmentation of financial clients between advanced and emerging markets, we replicated a questionnaire[1] measuring attitudes toward financial affairs as used in the study of Fünfgeld and Wang (2009). We also include questions that collect information about participants' psychological biases, personality traits, investment behavior, socio-demographic backgrounds and saving motives. In this study, however, we focus only on the first seventeen statements regarding financial attitudes and personal characteristics.

The questionnaire was translated into Vietnamese and back-translated into English by two independent translators with an economic background. The translated questionnaire was then pretested with a sample of 130 undergraduate students in an attempt to improve the format of the questions before studying the general population.

Sample description

The questionnaires were distributed to Vietnamese financial customers in

[1]The original questionnaire consists of 17 five-point-Likert-scale statements ranging from "1=absolutely" to "5=not at all"

paper-and-pencil format between March and May of 2016. A total of 3,480 respondents were recruited from three sources: 35% participants (n=1,220) were customers at ten security brokerage firms and commercial banks, together with participants in financial training workshops organized by the companies. The second source consists of 1,150 participants (33%) who were studying finance at universities. The third source was through a combination of the "quota and snowball sampling procedure" (Vogt and Johnson 2011) to avoid a financial literacy bias. A total of 1,110 respondents (31%) without a financial educational background was identified in this source.

In most cases, the survey was conducted at respondents' places and collected by the researchers within two weeks after distribution. Of the total sample of 3,144 questionnaires returned, 448 were omitted due to too few answered questions (80% as a cutting point) and response differences between two pairs of identical items[2] that were also used to discard unreliable data. As a result, a sample of 2,706 valid questionnaires was used for the further analysis (77%).

Participants

Table 1 summarizes demographic characteristics of Vietnamese and Swiss samples including gender, age, education from this analysis. It shows that the age of Vietnamese participants ranges from 18 to 85 years and 90% of them falls into the category of 18-35 years old. Sixty-four percent of the participants are female. The percentage of participants with university degrees is 51% whereas the percentage of people with only high school diplomas is 43%. Only a small number of individuals did not attend primary schools or attended but failed to obtain certificates (1%), reflecting the relatively high rate of literacy among the Vietnamese adult population, estimated at 98% in 2015 (UNESCO 2017). Moreover, 21.2% participants are non-students whereas 74.4% participants are full-time or external students at university or institutions of higher education. As compared to the Swiss study, fewer participants with apprenticeship certificates (3% in Vietnam vs. 34% in Switzerland where this type of degree is very common due to differences in the education system) and more university students participated in the survey in Vietnam (74.4% in Vietnam vs. 11% in Switzerland). However, the marked contrast toward demographic characteristics between two samples can be mitigated when considering respondents' working experience because the majority of the Vietnamese respondents were working simultaneously at the time of the survey. More specifically, only 33% participants were un-

[2]In this survey, two of the first 17 five-point-Likert-scale statements towards financial attitudes were repeated twice. Subjects were excluded from the dataset if providing different answers to those questions.

employed, retired people and homemakers, whereas 71% participants (70% of those aged below 35 years) were working full-time, part-time and self-employed (the sum is higher than 100% because they could choose more than one option in their employment statuses)[3]. Additionally, about half of the participants (41% of those who aged below 35 years)[3] earn more than 500 EUR per month (equivalent to 6,000 EUR annually), which is three times higher than the average Vietnamese annual income per person (WorldBank 2017). Hence, our dataset could be utilized to reflect financial attitudes and behavior of the young and educated Vietnamese population, which has been increased considerably over the last decade. This population is also very comparable to the Swiss sample regarding finance-related experience.

[3]Additional information is available upon request

Table 1: Demographic characteristics of Vietnamese and Swiss samples

	Vietnamese sample		Swiss sample	
	N	%	N	%
Gender				
Male	915	34%	632	49%
Female	1,741	64%	650	51%
Missing	50	2%		
Age				
Age 18-35	2,444	90%	427	33%
Age 36-50	182	7%	472	37%
Age 51-65	42	2%	283	22%
Age 66-85	5	0%	66	5%
Missing	33	1%	34	3%
Education				
Less than high school	31	1%	65	5%
Vocational school diploma	93	3%	433	34%
High school diploma	1,167	43%	186	15%
University degree	1,375	51%	598	47%
Missing	40	1%		
Students				
Non-students	573	21.2%	1,147	89%
University students	2,015	74.4%	135	11%
-Full-time university students	1,208	44.6%		
-External university students	807	29.8%		
Missing	118	1%		
Total	2,706	100%	1,282	100%

4 Results

4.1 *Findings of the Vietnamese study*

4.1.1 *Exploratory data analysis techniques*

To ensure a comparability between the results of our study and Fünfgeld and Wang (2009), we strictly follow their methodological framework. We begin our analysis by conducting the exploratory factor analysis based on the principal component analysis together with the Varimax rotation technique to extract underlying constructs of financial clients' attitudes and behavior

in Vietnam. Afterwards, the outputs of factor analysis are submitted to the Cronbach alpha inter-reliability test for variables to be reduced to maximize alpha scores (Hair et al. 1998).

By empirically classifying 2,706 respondents, the analysis procedure initially generated five factors, which can explain 51.5% of the total variance. Before coming up with the five-factor solution for the Vietnamese sample, information on Kaiser's criterion and Scree plots were taken into account because those techniques are essential for technically determining the number of extracted factors (Field 2005). As the requirement of the eigenvalues is greater than 1, Kaiser's criterion and Scree test suggested that the solution of five factors is the best in this case. Table 2 provides a summary of the five dimensions, item correlations (factor loading) and Cronbach alpha scores. The ranking of factor 1 to 5 of both samples reflects the declining eigenvalues.

In the following, we describe findings of the survey replication for Vietnamese financial consumers in comparison with those of the Swiss sample:

Number of eligible factors

The results of Vietnamese participants show lower Cronbach alpha coefficients in most constructs as compared to those of Swiss people. Notably, two of the five factors (factor 4 and factor 5) fail to meet a minimum Cronbach alpha value of 0.5 which is adequate in research for the use of a scale (Aron and Aron 1994). However, since it is a two-item scale with Spearman-Brown correlation of 0.357, which is significant at 1%, the factor 4 of the Vietnamese dataset can be treated as eligible for analysis (Eisinga et al. 2013). Therefore, despite the same extracted five factors through exploratory factor analysis, the Vietnamese sample presents four components within satisfactory standards of reliability, which justify further investigation.

Table 2: Comparison of factor composites between Vietnamese and Swiss samples

	Swiss sample (N = 1,282)			**Vietnamese sample** (N = 2,706)	
		Factor loadings			Factor loadings
Item	*Factor 1: Anxiety (α = 0.648)*		Item	*Factor 2: Anxiety (α = 0.558)*	
1	I get unsure by the lingo of financial experts	0.567	1		0.517
2	I am anxious about financial and money affairs	0.701	2		0.580
3	I tend to postpone financial decisions	0.679	3		0.724
4	After making a decision, I am anxious about financial & money affairs	0.713	4		0.713
	Factor 2: Interest in financial issues (α = 0.711)			*Factor 3: Interest in financial issues (α = 0.639)*	
5	I read the business section of the newspaper attentively	0.811	5		0.838
6	I like to join conversations about financial matters	0.826	6		0.846
			7		0.507
	Factor 3: Intuitive decisions (α = 0.604)			*Factor 1: Intuitive decisions & Free-spending (α = 0.644)*	
7	I compare and calculate risks	-0.537	8		0.680
8	Even on large purchases, I tend to spend spontaneously	0.629	11		0.633
9	I enjoy reading about the results of product comparison	-0.545	14		0.699
10	I do not complain very often, even if I have a reason to do so	0.508	16		0.692
11	At the end of the day, I decide intuitively in financial affairs	0.595			
	Factor 5: Free-spending (α = 0.611)			*Factor 5: Intuitive decisions & Free-spending (α = 0.438)*	
14	I spend money when I am unhappy or frustrated	0.623	9		0.704
15	Special offers can entice me into buying	0.705	15		0.628
16	I enjoy spending money more than saving	0.515	17		0.451
	Factor 4: Needs for precautionary saving (α = 0.501)			*Factor 4: Needs for precautionary savings (α = 0.490)*	
12	I find it hard not to have some money away for a rainy day	0.767	12		0.768
13	To care for the future is essential for me	0.706	13		0.669

Notes: 1. The ranking of factor 1 to 5 of both samples reflects the declining eigenvalues. 2. The sequence of Items describes orders in the questionnaire. 3. Factor loadings indicate the correlations of the included items within the factor after the Varimax orthogonal rotation. 4. α= value of Cronbach's alpha (based on standardized items) for the corresponding factor. 5. Factor 5 of the Vietnamese sample (including Item 9, 15 and 17) was not used for further analysis due to its low Cronbach's alpha score (a value of 0.5 as acceptable). 6. Item 10 "I do not complain very often, even if I have a reason to do so" of the Vietnamese sample and Item 17 "I feel annoyed when things do not go my way" of the Swiss have been deleted due to their lower loading scores.

Factor compositions

Table 2 also illustrates similarities in factor composites between Vietnamese and Swiss samples. It shows that two dimensions measuring the "Anxiety" and "Needs for precautionary savings" attitudes of Swiss consumers share the same composites with those of Vietnamese consumers. The "Interest in financial issues" construct of the Swiss sample is reflected in the outcomes of the Vietnamese dataset, but it slightly differs in the content as there is one more item (Item 7: "I compare and calculate risks") assigned to the corresponding dimension of the Vietnamese sample. This can indicate that Vietnamese participants show their interest towards financial matters by not only reading financial news or participating in financial conversations but also applying impersonal analysis to their decisions. Additionally, unlike Swiss consumers where impulsive spending and intuitive decision making are two distinct factors, for Vietnamese respondents, these two dimensions tend to be correlated, resulting in one factor labeled as "Intuitive decisions & Free-spending" (Factor 1 with the Cronbach alpha score of 0.644 in Table 2). In sum, there are remarkable similarities in the factor composites between the two countries even though they entirely differ in cultural, economic and socio-political dimensions. This suggests that scales based on personal financial attitudes and behavior in the study of Fünfgeld and Wang (2009) can be extended to other countries, including emerging markets.

Table 3: Comparison of factor composites between Vietnamese unweighted and weighted samples

	Vietnamese unweighted sample (N=2,706)			**Vietnamese weighted sample** (N=2,706)	
		Factor loadings			Factor loadings
Item	*Factor 1: Anxiety (α= 0.558)*		Item	*Factor 1: Anxiety (α= 0.438)*	
1	I get unsure by the lingo of financial experts	0.517	1		0.620
2	I am anxious about financial and money affairs	0.580			
3	I tend to postpone financial decisions	0.724	3		0.721
4	After making financial decisions, I am anxious about financial and money affairs	0.713	4		0.737
	Factor 2: Interest in financial matters (α=0.639)			*Factor 2: Interest in financial matters (α= 0.640)*	
5	I read the business section of the newspaper attentively	0.838	5		0.774
6	I like to join conversations about financial matters	0.846	6		0.733
7	I compare and calculate risks	0.507	7		0.606
	Factor 3: Intuitive decisions & Free-spending (α= 0.644)			*Factor 3: Intuitive decisions & free-spending (α= 0.644)*	
8	Even on large purchases, I tend to spend spontaneously	0.680	8		0.629
11	At the end of the day, I decide intuitively in financial affairs	0.633	11		0.698
14	I spend money when I am unhappy or frustrated	0.699	14		0.693
16	I enjoy spending money more than saving	0.692	16		0.709
	Factor 4: Intuitive decisions & Free-spending ((α=0.438)			*Factor 4: Intuitive decisions & free-spending (α= 0.600)*	
9	I enjoy reading about the results of product comparison	0.704	9		0.668
15	I do not complain very often, even if I have a reason to do so	0.627	15		0.622
17	I feel annoyed when things do not go on my way	0.451	10		0.578
	Factor 5: Needs for precautionary savings (α= 0.490)			*Factor 5: Needs for precautionary savings (α= 0.570)*	
12	I find it hard not to have some money away for a rainy day	0.727	12		0.690
13	To care for the future is essential for me	0.615	13		0.594
			2		0.540
			17		0.643

Notes: 1. The sequence of Items describes orders in the questionnaire. 2. Factor loadings indicate the correlations of the included items within the factor after the Varimax orthogonal rotation. 3. α = value of Cronbach's alpha (based on standardized items) for the corresponding factor. 4. Item 10 of the Vietnamese unweighted sample has been deleted due to its lower loading scores.

4.1.2 Robustness of Factor composites

One could argue that differences in socio-demographic distributions between two countries can affect the outcome of our replicate study. To address the issue, we assign weighting adjustment scores based on three dimensions: gender, age and students to each survey participant of the Vietnamese sample[4]. The results of the Vietnamese weighted sample are summarized in Table 3 and show that the five-factor solution and its factor composites are similar between the weighted and un-weighted samples. In particular, the "Interest in financial matters" and "Intuitive decisions and free-spending" constructs of the weighted sample (Factor 2 and 3 respectively) have the same composites with those of the unweighted sample. The remaining constructs of the weighted sample consist of main items as reflected in those of the unweighted sample. For instance, out 3 of 4 items are loaded in the same "Anxiety" construct (Factor 1) and two main items of "Intuitive decisions and Free-spending" and "Needs for precautionary savings" (Factor 4 and 5 respectively) can be found in those of the weighted sample. We, therefore, conclude that differences in sample sizes and socio-demographic distributions are not a severe problem and thus we will continue the analysis with the unweighted sample.

4.1.3 *Cluster analysis*

In this section, we study profiles of financial clients by applying cluster analysis to the Vietnamese sample and compare them with those of Swiss participants to examine whether similar patterns of segments of individual consumers emerge across nations and if not, whether any link might exist between cross-national variations and economic and cultural traits.

When making a cross-country comparison, it is important to build parallel constructs as the segmentation basis for classifying financial consumers. Despite similar results, the Vietnamese sample is slightly different from the Swiss sample regarding the factor components. To keep consistency, we decompose the "Intuitive decisions & Free-spending" factor of the Vietnamese sample into two elements with two items each. In this way, the first two items (Item 8 and 11) add their values to the new feature, labeled as "Intuitive decisions" and the remaining (Item 14 and 16) build up the "Free-spending" construct. Regarding the "Interest in financial issues" factor, we keep the original construct with only two items (Item 5 and 6), as validated in the study of Swiss investors. It means that the item "I compare and calculate risks" did not account for that factor. Similarly, the new setting of segment

[4]See Table A1 for more details.

variables applies to the Swiss dataset before a comparison of cluster analysis between two samples.

To compute factor loading scores as variables for cluster analysis, we use the same techniques as presented in the study of Fünfgeld and Wang (2009). The overall score of each factor is calculated by taking average scores for included items within this factor. In this way, the factor has the same scale (1-5) and does not need to be standardized. In the next step, we have to examine correlations of the five factors as cluster analysis requires independent variables. By looking at the set of variables, we suspect that there are some high correlations in our dataset, particularly that of the two decomposed factors (Intuitive decisions & Free-spending) since they originate in the same factor of the principal component analysis. However, none of the correlations exceeds 0.5 (for more details, see Table A2 of Appendix); and thus, the independent criterion is satisfied (Cohen 1988).

In the procedure of cluster analysis, we employ a "two-stage mixing" approach which emphasizes both hierarchical and partitional methods. Firstly, the hierarchical clustering method with Ward's linkage and squared Euclidean distances has been conducted to identify the number of clusters and define group centroids. Previously, outlying and missing observations in scores of five factors of both datasets are examined to avoid distorting the results of Ward methods. Using Hair et al. (1998) method, 20 extreme cases are detected in the Vietnamese sample for their scores that deviate more than three standard deviations from the mean. All cases are then eliminated from the analysis. For two missing values that exist in each subject pool, we conduct the "mean imputation" method by substituting them with mean values of the corresponding components. The next step is to provide cluster membership by using centroids of Ward's solution as starting seeds in the partitional clustering K-means method. Such agglomerative method can be demonstrated superior to the standard K-means with random centers regarding cluster recovery (Punj and Stewart 1983) and the likelihood of the global optimum to be reached (Bacher et al. 2010).

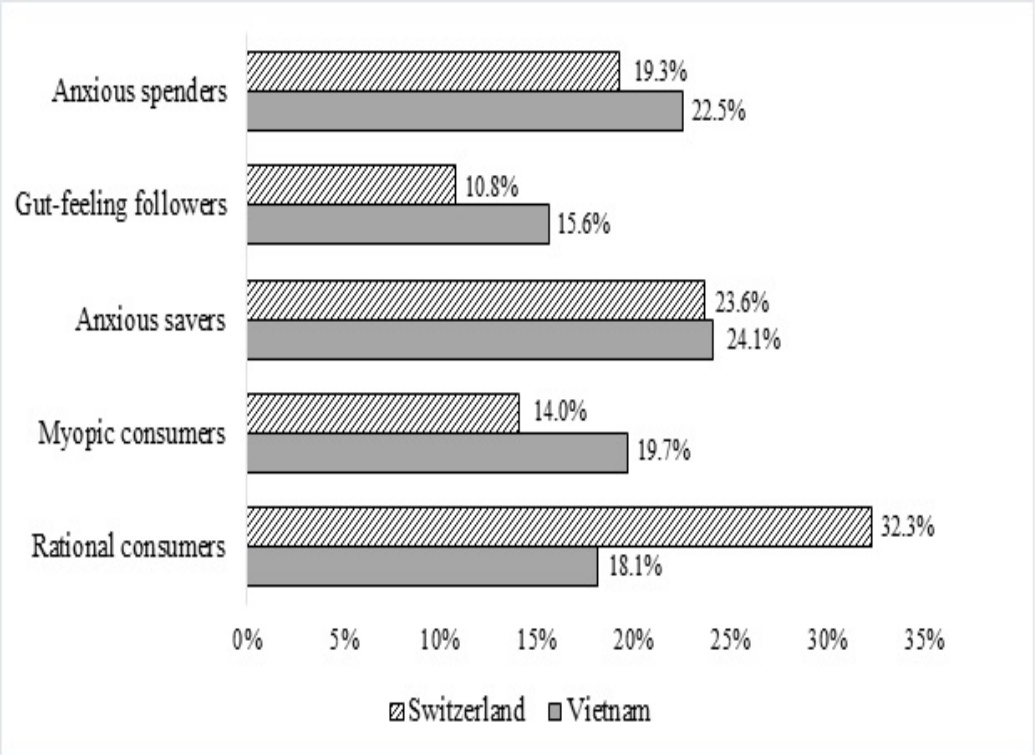
In the following sections, we discuss the profile of Vietnamese individual segments in comparison with those from Switzerland.

Cluster distribution

Figure 1 displays the group distributions of the two samples. The five groups of retail financial clients were labeled to stress the similarity with the classification scheme as suggested by Fünfgeld and Wang (2009). The results of the Vietnamese sample highlight that the *Anxious savers* make up the highest proportion of participants, similar in size to the Swiss sample (24.1% and 23.6% respectively). Further, there are more *Myopic consumers*, *Gut-feeling followers*, and *Anxious spenders* in the Vietnamese sample than

in the Swiss sample. By contrast, the proportion of Vietnamese participants who are classified as *Rational consumers* is much lower than that of Swiss participants in the corresponding cluster.

Figure 1: Comparison of Cluster distribution between two countries



Profile analysis

In this section, we take a closer look at cluster membership of the Vietnamese sample in a comparative perspective. We hypothesize that if there are more similarities than dissimilarities in segments of financial clients between two samples, we may deduce that scales measuring individuals' financial attitudes and behavior as proposed by Fünfgeld and Wang (2009) can be universally used for classifying consumers in the international market. Table 4 reports the detailed interpretation of each cluster of the two samples.

Table 4: Profiling of financial clients (based on final cluster centers) in a comparative perspective

		Rational consumers	Myopic consumers	Anxious savers	Gut-feeling followers	Anxious spenders
Vietnamese sample (N= 2,686)	Anxiety	Medium (-0.24)	**Low** (-0.55)	**High** (0.31)	**Medium** (-0.13)	**High** (0.44)
	Interest in financial issues	**High** (1.39)	Medium (-0.08)	**Low** (-0.62)	**Low** (-0.56)	Medium (-0.01)
	Intuitive decisions	Low (-0.43)	Low (-0.39)	**Low** (-0.58)	**High** (0.44)	**High** (0.96)
	Needs for precautionary savings	**High** (0.51)	**Low** (-0.96)	**High** (0.69)	**Low** (-1.12)	**High** (0.51)
	Free spending	**Low** (-0.49)	Low (-0.59)	**Low** (-0.61)	**High** (0.75)	**High** (1.03)
Swiss sample (N= 1,282)	Anxiety	Low (-0.43)	**Low** (-0.61)	**High** (0.28)	**Medium** (0.13)	**High** (0.76)
	Interest in financial issues	**High** (0.90)	High (0.64)	**Low** (-0.74)	**Low** (-1.04)	Low (-0.49)
	Intuitive decisions	Medium (-0.17)	Medium (-0.08)	**Low** (-0.56)	**High** (0.64)	**High** (0.69)
	Needs for precautionary savings	**High** (0.51)	**Low** (-1.40)	**High** (0.50)	**Low** (-1.56)	**High** (0.43)
	Free-spending	**Low** (-0.47)	Medium (0.01)	**Low** (-0.46)	**High** (0.67)	**High** (0.97)

Notes: The table gives an overview of the clusters as related to five factors of the two samples in a comparable setting. Numbers in parentheses refer to the value of the factor converted to Z-scores. Texts in bold show similarities of the dimensions within the specific cluster of the two samples.

- ***Cluster 1–Rational consumers***: members of this segment show strong interest in financial matters and high preference for saving over impulsive spending. However, Vietnamese participants in this group slightly differ from those of the Swiss sample regarding anxiety and intuitive decisions. In particular, they appear to be more anxious and less intuitive in their decision-making compared with Swiss.

- ***Cluster 2–Myopic customers***: members of both samples in this cluster have common features regarding anxiety and savings: they feel at ease with financial affairs and show low needs for precautionary savings. However, they significantly differ from each other in the remaining dimensions. More specifically, Vietnamese people tend to be less interested in financial matters, less intuitive in financial decision-making and less prone to spontaneous spending than Swiss.

- ***Cluster 3 –Anxious savers***: They tend to be highly anxious and in need of precautionary saving. They are less likely to be involved in impulsive spending and more analytical in financial decision-making despite their low interest in financial matters. In general, Vietnamese and Swiss customers in this cluster are almost indistinguishable.

- ***Cluster 4–Gut-feeling followers***: Members of this group are characterized by uninterestedness in financial matters, intuitive decision-making styles and low needs for precautionary saving. Moreover, they are more likely to spend spontaneously. As compared to Swiss gut-feeling followers, Vietnamese share the same features in most dimensions.

- ***Cluster 5–Anxious spenders***: Members of this group show very complex attitudes. They feel insecure with financial matters and tend to make financial decisions intuitively. Further, they enjoy spending and perceive it as a means of improving their mood, but at the same time they feel a high need of saving. Such complex attitudes are very similar between Vietnamese and Swiss participants in this cluster.

The above results highlight that the segmentation of Vietnamese financial consumers is remarkably similar within each cluster as compared to that of Switzerland, especially for the clusters of Anxious savers, Gut-feeling followers and Anxious spenders. Therefore, the scales based on individual "day-to-day" financial attitudes and behavior can be used as segmentation variables for identifying similar market segments across different countries.

Table 5: Socio-demographic characteristics of the comparable five factors of Vietnamese and Swiss samples

	Whole sample		Rational consumers		Myopic consumers		Anxious savers		Gut-feeling followers		Anxious spenders		χ^2
	N	%	N	%	N	%	N	%	N	%	N	%	
Gender													
Male	905 (632)	33.7 (49.3)	*236 (277)*	*26.1 (43.8)*	189 *(119)*	20.9 *(18.8)*	*158 (112)*	*17.5 (17.7)*	129 (62)	14.3 (9.8)	193 *(62)*	21.3 *(9.8)*	$\chi^2(8)$=79.38***
Female	1,731 (650)	64.4 (50.7)	*240 (137)*	*13.9 (21.1)*	328 *(61)*	18.9 *(9.4)*	*477 (191)*	*27.6 (29.4)*	283 (76)	16.3 (11.7)	403 *(185)*	23.3 *(28.5)*	($\chi^2(4)$=149.08***)
χ^2			$\chi^2(2)$=60.23***		$\chi^2(2)$=1.59		$\chi^2(2)$=33.14***		$\chi^2(2)$=1.98		$\chi^2(2)$=2.53		
			($\chi^2(1)$=75.87***)		($\chi^2(1)$=23.68***)		($\chi^2(1)$=24.15***)		($\chi^2(1)$=1.18)		($\chi^2(1)$=71.66***)		
Age													
18-35	2,427 (427)	90.4 (33.3)	435 *(115)*	17.9 *(26.9)*	*455 (68)*	*18.7 (15.9)*	583 *(110)*	24.0 *(25.8)*	371 *(55)*	15.3 *(12.9)*	*583* (79)	*24.0* (18.5)	$\chi^2(12)$=47.41***
36-50	180 (472)	6.7 (36.8)	42 *(152)*	23.3 *(32.2)*	*47 (71)*	*26.1 (15.0)*	44 *(95)*	24.4 *(20.1)*	31 *(61)*	17.2 *(12.9)*	*16* (93)	*8.9* (19.7)	($\chi^2(16)$=45.08***)
51-85	46 (349)	1.7 (27.2)	4 *(140)*	8.7 *(40.1)*	*16 (40)*	*34.8 (11.5)*	13 *(82)*	28.3 *(23.5)*	11 *(22)*	23.9 *(6.3)*	*2* (65)	*4.3* (18.6)	
χ^2			$\chi^2(3)$=6.11		$\chi^2(3)$=15.05***		$\chi^2(3)$=0.61		$\chi^2(3)$=3.74		$\chi^2(3)$=34.45***		
			($\chi^2(3)$=17.51***)		($\chi^2(3)$=7.04*)		($\chi^2(3)$=14.63***)		($\chi^2(3)$=15.61***)		($\chi^2(3)$=2.56)		
Education													
Secondary school or less	30 (65)	1.1 (5.1)	*5* (18)	*16.7* (27.7)	*5 (3)*	*16.7 (4.6)*	*9* (19)	*30.0* (29.2)	6 (9)	20.0 (13.8)	*5 (16)*	*16.7 (24.6)*	$\chi^2(16)$=79.47***
High school diploma	1,161 (186)	43.2 (14.5)	*162* (57)	*14.0* (30.6)	*178 (33)*	*15.3 (17.7)*	*307* (36)	*26.4* (19.4)	202 (18)	17.4 (9.7)	*312 (42)*	*26.9 (22.6)*	($\chi^2(12)$=20.13*)
Apprenticeship	92 (433)	3.4 (33.8)	*24* (130)	*26.1* (30.0)	*15 (58)*	*16.3 (13.4)*	*27* (102)	*29.3* (23.6)	16 (47)	17.4 (10.9)	*10 (96)*	*10.9 (22.2)*	
University degree	1,363 (598)	50.7 (46.6)	*286* (209)	*21.0* (34.9)	*320 (86)*	*23.5 (14.4)*	*294* (146)	*21.6* (24.4)	190 (64)	13.9 (10.7)	*273 (93)*	*20.0 (15.6)*	
χ^2			$\chi^2(4)$=26.36***		$\chi^2(4)$=27.90***		$\chi^2(4)$=10.23**		$\chi^2(4)$=6.37		$\chi^2(4)$=28.82***		
			($\chi^2(3)$=3.81)		($\chi^2(3)$=7.10*)		($\chi^2(3)$=3.22)		(X2(3)=0.88)		($\chi^2(3)$=10.16**)		
Students													
University students	2,001 (135)	74.5 (10.5)	369 *(55)*	18.4 *(40.7)*	*348 (32)*	*17.4 (23.7)*	487 *(22)*	24.3 *(16.3)*	306 (9)	15.3 (6.7)	*491 (17)*	*24.5 (12.6)*	$\chi^2(8)$=37.48***
Non-students	569 (1,147)	21.2 (89.5)	97 *(359)*	17.0 *(31.3)*	*149 (148)*	*26.2 (12.9)*	132 *(281)*	23.2 *(24.5)*	93 (129)	16.3 (11.2)	*98 (230)*	*17.2 (20.1)*	($\chi^2(4)$=22.65***)
χ^2			$\chi^2(2)$=0.58		$\chi^2(2)$=25.53***		$\chi^2(2)$=0.32		$\chi^2(2)$=0.93		$\chi^2(2)$=19.95***		
			($\chi^2(1)$=4.93**)		($\chi^2(1)$=11.67***)		($\chi^2(1)$=4.50**)		($\chi^2(1)$=2.64)		($\chi^2(1)$=4.32**)		
Whole sample	2,686 (1,282)	100 (100)	487 (414)	18.1 (32.3)	528 (180)	19.7 (14.0)	647 (303)	24.1 (23.6)	420 (138)	15.6 (10.8)	604 (247)	22.5 (19.3)	

Notes: Numbers in parentheses refer to the statistics of the Swiss sample in the relative criteria. χ^2 is tested based on "priori" probabilities to the study sample. Figures in italics show significant differences between the groups within the specific cluster. *Significant at 10%, ** Significant at 5%, ***Significant at 1%. .

Table 6: Comparison of OLS regressions for socio-demographic characteristics of the five clusters between the two samples

Variables	Anxiety		Interest in financial issues		Intuitive decisions		Needs for precautionary savings		Free-spending	
	Vietnam	Swiss	Vietnam	Swiss	Vietnam	Swiss	Vietnam	Swiss	Vietnam	Swiss
Gender (Male = 0)	0.045*	0.406***	-0.335***	-0.678***	-0.061*	0.060	0.055	0.280***	0.06	0.429***
	(0.03)	(0.05)	(0.04)	(0.06)	(0.03)	(0.06)	(0.04)	(0.07)	(0.04)	(0.05)
Age										
18-35										
36-50	-0.074	-0.043	0.140*	0.351***	-0.159**	0.149**	-0.170**	0.058	-0.243***	-0.096
	(0.06)	(0.06)	(0.08)	(0.08)	(0.07)	(0.08)	(0.08)	(0.09)	(0.08)	(0.07)
51-85	-0.061	0.046	-0.399**	0.549***	-0.368**	0.147*	-0.571***	0.321***	-0.334***	-0.293***
	(0.12)	(0.07)	(0.16)	(0.08)	(0.15)	(0.08)	(0.17)	(0.09)	(0.13)	(0.07)
Education										
Secondary school or less	0.357**	0.159	-0.272	0.069	0.084	0.048	0.018	0.017	0.043	-0.211
	(0.17)	(0.13)	(0.22)	(0.16)	(0.15)	(0.17)	(0.18)	(0.18)	(0.18)	(0.15)
High school diploma										
Apprenticeship	0.005	-0.094	0.267**	0.188*	-0.355***	0.013	0.068	0.025	-0.470***	-0.281***
	(0.08)	(0.08)	(0.13)	(0.10)	(0.10)	(0.10)	(0.10)	(0.11)	(0.12)	(0.09)
University degree	-0.078***	-0.169**	0.245***	0.238**	-0.104***	-0.211**	0.046	-0.008	-0.207***	-0.250***
	(0.03)	(0.08)	(0.04)	(0.10)	(0.04)	(0.10)	(0.04)	(0.12)	(0.04)	(0.09)
Students (non-students = 0)	0.062*	-0.368***	0.127**	0.613***	0.01	-0.241**	0.171***	0.127	-0.054	-0.056
	(0.04)	(0.08)	(0.05)	(0.11)	(0.04)	(0.11)	(0.05)	(0.12)	(0.05)	(0.10)
Constant	3.156***	2.462***	2.808***	2.890***	2.465***	2.885***	3.866***	3.724***	2.618***	2.737***
	(0.05)	(0.08)	(0.06)	(0.11)	(0.06)	(0.10)	(0.06)	(0.12)	(0.06)	(0.09)
R-squared	0.016	0.095	0.058	0.135	0.018	0.025	0.02	0.03	0.026	0.074
F	5.251	19.43	20.65	30.99	6.85	4.652	5.74	5.67	9.751	15.12

Notes: n = 2,686 (Vietnam) and 1,282 (Switzerland). The robust standard error is in brackets; Multicollinearity is tested by Variance inflation factors (VIF); *Significant at 10%, ** Significant at 5%, ***Significant at 1%.

4.1.4 *Cluster membership and socio-demographic variables*

Table 5 reports descriptive statistics of socio-demographic variables for each group in Vietnamese and Swiss samples respectively. We find that socio-demographic variables including gender, age, and education show significant differences in their distributions over the five clusters of both samples (as shown by the Chi-squared test in the last column of Table 5). Within each investor group, the distribution of socio-demographic variables varies across countries.

Gender

The first two rows of Table 5 show the distribution of consumer types within each gender. For both Vietnamese and Swiss samples, female participants are more likely to belong to the *Anxious savers* (27.6% female vs. 17.5% male in the Vietnamese sample; 29.4% female vs. 17.7% male in the Swiss sample), but they are much less likely to be in the *Rational consumers* group (13.9% female vs. 26.1% male in the Vietnamese sample; 21.1% female vs. 43.8% male in the Swiss sample). Moreover, there are significant gender differences only for the Swiss sample regarding the *Anxious spenders* and *Myopic consumers* groups. Women are more likely to be in the *Anxious spenders* group, but they are less likely to appear in the *Myopic consumers* group. More specifically, 28.5% female participants are classified as *Anxious spenders* compared with only 9.8% male participants ($p < .01$, $\chi^2(1) = 71.66$), whereas only 9.4% female participants are considered as *Myopic consumers* compared with 18.8% male participants ($p < .01$, $\chi^2(1) = 23.68$).

The first two rows in Table 6 show gender differences in five factors. Both Swiss and Vietnamese women are less interested in financial matters than men. Swiss women tend to be more anxious, more in need of precautionary saving, and more subject to excessive spending (e.g. Anxious spenders), whereas Vietnamese women are only marginally more anxious than men, and not different from men in attitudes towards saving and spending. We find no significant gender difference for both countries regarding intuitive decision-making. These results can suggest that women lack intrinsic interest in financial matters in both countries. It is interesting to observe a more dominant gender difference in Switzerland, whereas Vietnamese women and men are much more similar in most dimensions. Such patterns might be traced back to a higher masculinity culture in Switzerland where men and women are expected to play different social roles, whereas Vietnam has a more feminine oriented culture where gender roles are less distinguished (Hofstede et al. 2010).

Age

Table 5 shows that only group differences in the *Myopic consumers* and

Anxious spenders clusters of the Vietnamese sample are significant (p <.01, $\chi^2(3) = 15.05$ and $\chi^2(3) = 34.45$, respectively). Within the *Anxious spenders* group, older participants are under-represented (respectively 4.3% and 8.9% participants in the age-groups of the 36-50 and the over 50 years old vs. 24% of those aged below 35 years). Conversely, they are more likely to be found in the *Myopic consumers* segment than the younger (26.1% and 34.8% participants in the age groups of the 36-50 and the over 50 years old respectively vs. 18.7% of those aged under 35 years). Within these age groups, Vietnamese people are less likely to make intuitive financial decisions and impulsive spending but also have a lower need for precautionary saving than the younger participants as reflected in the OLS regression models (Table 6). This can imply that Vietnamese older participants seem to take more control of their financial decisions than the younger ones, but they also show a "Life is at now" attitude. Due to too few participants in the oldest group of the Vietnamese sample, we are cautious to make general conclusions based on the current results.

In contrast, we observe significant age effects for each cluster in the Swiss sample, except for the *Anxious spenders* group (p>.10, $\chi^2(3)= 2.56$). There is a high proportion of people aged over 50 years in the *Rational consumers* (40.1% participants in the over-50 group vs. 32.2% and 26.9% participants aged 36-50 and below-35 respectively). Swiss participants in the above-50 group tend to care more for saving and show high interest in financial matters, but they are less likely to spend spontaneously. They are less likely to be found in the *Gut-feeling followers* and *Myopic consumers* segments than the younger ones for the same reasons. More specifically, only 11.5% participants in the over-50 group are classified as *Myopic consumers* compared with respectively 15% and 15.9% participants in the age-groups of 35-50 and below 35. Similarly, only 6.3% participants aged over 50 are considered as *Gut-feeling followers* compared with 12.9% (×2) participants in the other age groups. Within the *Anxious savers*, however, the results are mixed between the older and younger members. Notably, only 20.1% participants aged between 36 and 50 years are categorized as *Anxious savers* compared with respectively 23.5% and 25.8% people in the over-50 and the under-35 groups. Altogether, Swiss older participants seem to be better in their financial management than the younger, especially those aged below-35.

Education

With regards to education, Table 5 shows significant group differences across clusters for Vietnamese participants, except for the *Gut-feeling followers* segment (p>.10, $\chi^2(4)= 6.37$). Participants who achieved apprenticeship certificates and university degrees as the highest level of education are more likely to be in the *Rational consumers* group (47.1%). As reflected in

OLS regression models (Table 6), those members on average are less likely to make intuitive financial decisions and excess spending, and they are interested in financial matters. Besides, participants with university degrees show a lower level of anxiety with financial affairs. These results may also explain the tendency that participants with university degrees are less likely to belong to the *Anxious savers* (21.6% vs. 26.4% participants) and *Anxious spenders* (20.0% vs. 26.9% participants), whereas they are most frequently to be found in the *Myopic consumers* group (23.5% vs. 15.3% participants) compared with those who earned high school diplomas. On the other hand, members who are currently studying at a university are more likely to be found in the *Anxious spenders* (17.2% non-students vs. 24.5% students), but they are less likely to belong to the *Myopic consumers* group (17.4% students versus 26.2% non-students). The OLS regression models (Table 6) also capture such impressive results when showing that Vietnamese students tend to care more about saving and integrate into the financial world despite their high anxiety about it.

Education is not a significant predictor for cluster membership of our Swiss sample. Only the *Myopic customers* and *Anxious spenders* groups are significant ($p<.10$, $\chi^2(4)=7.10$ and $p<.05$, $\chi^2(4)= 10.16$ respectively). In particular, participants with higher levels of education are less likely to belong to the *Anxious spenders* group (eg. 15.6% university graduates vs. 22.6% of those with high school certificates), whereas those with lower levels of education, especially high school diplomas, are most often found to be in the *Myopic customers* cluster (eg. respectively 13.4% and 14.4% participants with apprenticeship and university graduates vs. 17.7% those with high school certificates). Additionally, there are more university students in *Rational consumers* and *Myopic consumers* segments than non-student customers, e.g., the students are interested in financial matters, less worried about financial matters and less prone to the intuitive style of financial decision-making. They are less likely to be in the groups of *Gut-feeling followers*, *Anxious savers* and *Anxious spenders* for the same reasons (Table 5).

Table 7: Summary of differences in the distribution of socio–demographic variables over the five cluster of both samples

			Rational consumers	Myopic consumers	Anxious savers	Gut-feeling followers	Anxious spenders
Gender		Vietnamese	-		+		
(Female)		Swiss	-	-	+		+
	< 35 years	Vietnamese		-			+
Age		Swiss	-	+	+	+	
	35-50	Vietnamese		+			-
		Swiss	+	-	-	+	
	51-85	Vietnamese		+			-
		Swiss	+	-	+	-	
Education	High school diploma or less	Vietnamese	-	-	+		+
		Swiss		+			+
	University degree	Vietnamese	+	+	-		-
		Swiss		-			-
	University students	Vietnamese		-			+
		Swiss	+	+	-		-

Notes: "−"/"+" denote the less/more likelihood of being in the corresponding cluster.

Table 7 summarizes the distribution of socio-demographic variables over the five groups. It shows that female participants in both countries are more likely to belong to *Anxious savers*, but they are less likely to be in the *Rational consumers* group. Swiss female participants are more often found in the *Anxious spenders* and *Myopic consumers* groups, whereas no gender differences are found in the Vietnamese sample. Further, there are more dissimilarities than similarities regarding age and education. Indeed, Vietnamese older participants are more likely to be in the *Myopic consumers*, which differs from Swiss people at the same ages. While university students are more likely to be characterized as *Rational consumers* in the overall Swiss sample, they are more likely to be in the *Anxious spenders* group of the Vietnamese sample. However, for both samples, participants with high school diplomas are more likely to be in the *Anxious spenders*, but those with university degrees are less likely to be in that group. Similarly, Vietnamese younger participants, especially the below-35 age group are classified as *Anxious spenders* whereas those of the Swiss sample are less likely to be in the *Rational consumers* group. Thus, these results can suggest that younger participants and those with a lower level of education in both countries seem to have some common characteristics, e.g., they are exposed to a higher risk of unreasonable financial management. Given the recent economic development of Vietnam, this is of course not surprising.

To summarize, we find that while the overall segmentation by financial attitudes is similar, the demographic factors that affect the positioning within this segmentation differs substantially between Switzerland and Vietnam. This suggests that customer segmentation by demographic characteristics is problematic when considering international customers and a more direct clustering, e.g. following the approach of Fünfgeld and Wang (2009) will work better.

4.2 *Robustness tests with the joint sample of Vietnamese and Swiss participants*

To have a robustness check of the findings on the validity of the five–dimension scales based on individual customers' self-reported financial attitudes (anxiety, interest in financial issues) and behavioral patterns (spending and saving), we apply a cluster analysis to the joint dataset as generated from Vietnamese and Swiss samples. Afterwards, we employ a Probit-regression with group membership as dependent variables, where the independent variable is a proxy for the nation and controlling for socio-demographic characteristics to examine whether economic and cultural traits influence the likelihood of

being in a particular cluster.

Table 8 reports profiling of financial clients in the joint sample. The results highlight that five clusters are identified and reveal key features, similar to those of the classification when we were analyzing data samples of those countries separately.

Table 8: Profiling of financial clients (based on final cluster centers) in the joint sample of Swiss and Vietnamese participants

	Rational consumers	Myopic consumers	Anxious savers	Gut-feeling followers	Anxious spenders
1. Anxiety	Low	Medium	High	Low	High
	(-0.54)	(-0.22)	(0.43)	(-0.32)	(0.56)
2. Interest in financial issues	High	Low	Low	Medium	Low
	(1.21)	(-0.33)	(-0.57)	(-0.04)	(-0.40)
3. Intuitive decisions	Medium	Low	Low	High	High
	(-0.15)	(-0.50)	(-0.67)	(1.05)	(0.67)
4. Needs for precautionary savings	High	Low	High	Low	High
	(0.46)	(-0.95)	(0.64)	(-1.17)	(0.54)
5. Free spending	Low	Low	Low	High	High
	(-0.41)	(-0.32)	(-0.71)	(0.91)	(0.92)
Number of participants	924	789	866	461	879
%	24%	20%	22%	12%	22%

Notes: n = 3,919 (after 49 extreme cases excluded). Numbers in parentheses refer to the value of the factor converted to Z-scores; The requirement for independent variables of the joint sample was tested by Pearson correlation matrix for the five attitudinal dimensions as presented in Table A3 of Appendix.

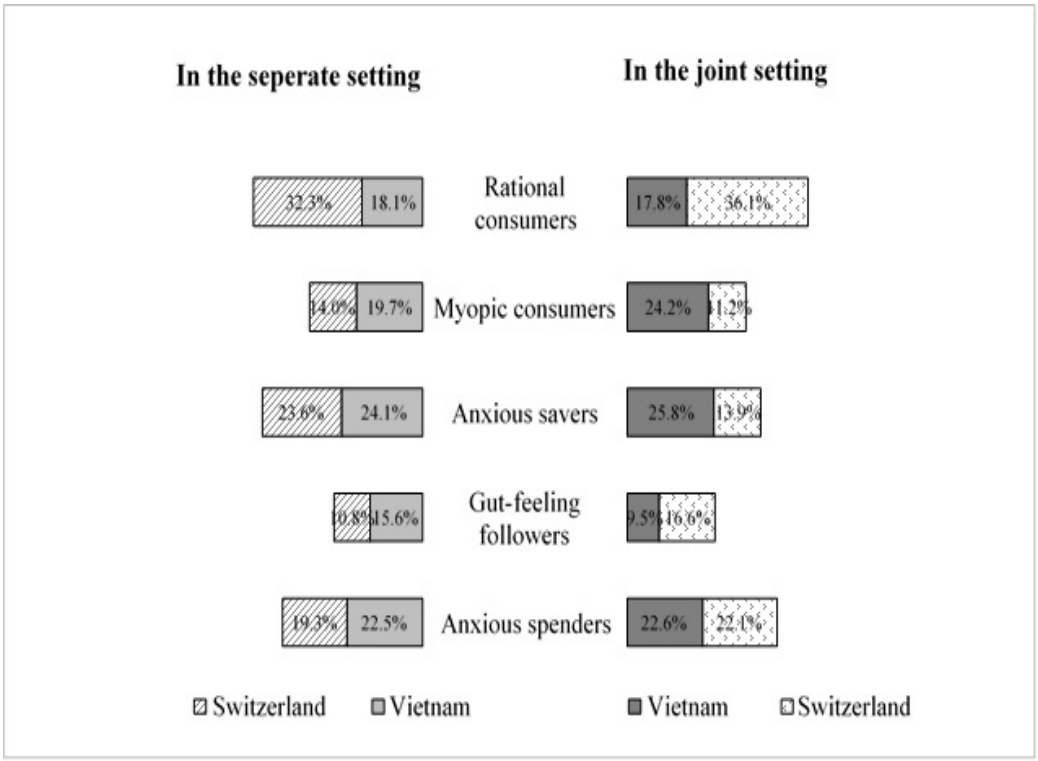
Table 9: Probit Regression of socio-demographics on cluster membership of the joint sample of Swiss and Vietnamese participants

VARIABLES	Rational consumers	Myopic customers	Anxious savers	Gut-feeling followers	Anxious spenders
Female	-0.675***	-0.236**	0.471***	-0.137	0.815***
	(0.08)	(0.10)	(0.10)	(0.09)	(0.09)
Nation (Switzerland = 0)	-0.611***	0.453***	0.566***	-0.347***	0.215**
	(0.08)	(0.10)	(0.10)	(0.10)	(0.09)
Female×Nation	0.230**	0.253**	-0.172	0.059	-0.701***
	(0.10)	(0.11)	(0.11)	(0.12)	(0.11)
Age					
18-35					
36-50	0.173**	-0.036	-0.065	0.028	-0.147*
	(0.08)	(0.08)	(0.08)	(0.09)	(0.08)
51-80	0.398***	-0.208*	0.031	-0.138	-0.260***
	(0.09)	(0.11)	(0.10)	(0.11)	(0.10)
Education					
Secondary school or less	0.295*	0.005	0.024	-0.196	-0.164
	(0.16)	(0.18)	(0.16)	(0.20)	(0.16)
High school diploma					
Apprenticeship	0.329***	-0.032	0.032	-0.006	-0.316***
	(0.09)	(0.10)	(0.09)	(0.10)	(0.09)
University degree	0.335***	0.07	-0.065	-0.076	-0.249***
	(0.06)	(0.06)	(0.06)	(0.07)	(0.06)
Students (Non-students = 0)	0.353***	-0.203***	-0.066	-0.094	0.001
	(0.07)	(0.07)	(0.07)	(0.09)	(0.07)
Constant	-0.535***	-1.039***	-1.331***	-0.811***	-0.889***
	(0.09)	(0.10)	(0.10)	(0.10)	(0.09)
Observations	3,734	3,734	3,734	3,734	3,734
Pseudo R^2	0.0846	0.0312	0.0349	0.0189	0.0326

Notes: n = 3,919 (after 49 extreme cases excluded). Numbers in parentheses refer to the value of the factor converted to Z-scores.

Table 9 displays the relationship between individual socio-demographic backgrounds and cluster membership. It can be seen that Vietnamese financial customers are more likely to be in the groups of *Myopic consumers, Anxious savers* and *Anxious spenders*, and less likely to be *Rational consumers* and *Gut-feeling followers*. Figure 2 presents the comparison of cluster distributions in separate samples as well as the joint sample. It seems that the general distribution of clusters is similar for both pooled and separate samples. In particular, the Swiss participants are more likely to be clustered as *Rational consumers*, and the Vietnamese are more likely to be classified as *Myopic consumers*. The between-country differences for the other three clusters are less robust.

Figure 2: Comparison of Cluster distribution between two countries



Regarding gender effect, Table 9 shows that Swiss female financial consumers are less likely to be in the *Rational consumers* and *Myopic consumers* but they are often found in the *Anxious savers* and *Anxious spenders* groups. This finding is in line with our findings when analyzing the two country samples separately. Adding an interaction effect between gender and nation into

our regression models, controlled by other socio-demographic variables, the gender difference is significantly smaller or diminished for the Vietnamese sample regarding the likelihood of *Rational consumers*, *Myopic consumers*, and *Anxious Spenders*, whereas women are more likely to be *Anxious savers* in both countries[5]. These results are consistent with our previous findings. The underlying reasons behind the cross-country differences of gender effects deserve further investigation.

Table 9 also provides information on age and education, indicating that older participants (aged over 35 years), people with university degrees and university students are most likely to belong to the *Rational consumers* group. This can imply that they are more efficient in financial management than younger and those with a lower level of education. In brief, these findings on the empirical relationship between socio-demographics and cluster membership support the results in section 4.1.3 and 4.1.4.

5 Discussion and Conclusion

Is it possible to find a classification for private investors regardless of their cultural and economic roots? So far the international marketing literature has not studied variables for segmenting individual consumers based on financial attitudes and behavior.

From a theoretical perspective, we provide empirical evidence on universal segmentation variables that reflect financial clients' psychological traits (anxiety, intuitive decision, and interest in financial matters) and attitudes toward saving and spending. It seems that our results, to some extent, are independent of cultural and economic differences, which supports the possibility of existence or emergence of a homogeneous global consumer culture (Cleveland and Laroche 2007). To our knowledge, this is the first cross-cultural study where finance-related attitudes and behavior are used as variables for segmenting financial customers. In this way, our research attempts to fill a gap in the literature.

From a practical viewpoint, our results can help financial advisors and practitioners to develop cross-national marketing strategies that take into consideration specific financial needs of each investor group to offer services more efficiently to them. In particular, financial advisors and practitioners should provide *Myopic consumers* with financial consultancy that emphasizes

[5]Based on the framework of interaction effects in non-linear models, proposed by Ai and Norton (2003) and Norton et al. (2004), we observe that interaction effects between gender and cluster membership can be interpreted similarly as signs of coefficients of the Probit regression models for most observations. These results are available upon request.

precautionary savings–the weakest point in their financial management. For example, realistic facts about expected consumptions after retirement and the need for retirement savings can raise their clients' awareness on retirement planning. In dealing with *Anxious savers*, financial advisors should stress the security aspect of financial services to mitigate their customers' anxiety. A clear decision making process, or as Thaler (1994) suggested, labels on products, can help customers feel more comfortable while making financial decisions. Regarding the *Gut-feeling customers* group, commitment devices or penalties to encourage saving and refrain clients from spontaneous spending might be helpful. Similar measures can also be applied to the *Anxious spenders* because both groups are common in terms of intuitive decisions and excessive spending. However, further support should be addressed to *Anxious spenders* as they show more complex attitudes towards financial matters than *Gut-feeling followers*. More specifically, they are in high need for precautionary savings but at the same time they take intuitive decisions and are prone to impulsive spending. They feel insecure about financial matters despite a high interest in the issue. Therefore, methods for keeping budgets, balancing saving and spending, and relieving anxiety could be suitable when communicating with this segment. Regarding the *Rational consumers* group, financial advisors have a much easier task because they demonstrate the most thoughtful and prudent financial behavior in the overall sample.

Our data also reveals that there is a substantial link between socio-demographic variables and cluster membership in both countries. Female, younger participants and those with a lower level of education in both countries have some common characteristics; they are exposed to a higher risk of unsound financial management. However, there was one interesting difference: greater attention should be given to female participants in Switzerland but not necessarily in Vietnam because in Switzerland, women seem to have more potential problems on controlling their financial decisions than in Vietnam. In general, these results confirm that socio-demographic variables still keep their validity as a basis for segmentation and can be taken into account when offering financial services globally. This gives further support to the usefulness on clustering dimensions in the international market setting.

Despite the contributions, a few limitations remain, which suggest some possibilities of further research. First, although we made efforts to increase the representativeness of our Vietnamese sample, there are still some deviations in its compositions as compared to the Swiss sample. Second, a four-cluster or six-cluster solution would perhaps be fit slightly better for Vietnamese respondents, but the comparison with the Swiss sample would have been more difficult. Last, there are small differences in the composition of the revealed dimensions based on participants' financial attitudes and

behavior between the two countries, mainly a mixture of intuitive decisions and free spending aspects. In general, however, the differences are surprisingly small, given the large discrepancies between the two countries. Future research may investigate more countries with comparable samples to further test the reliabilities and validities of the scales.

A Appendix

Table A1: Stratified samples and weighting scores

Groups	Swiss sample		Vietnamese sample		Weighting scores
	N	Percent	N	Percent	
Female students aged between 18 and 35 years	33	3%	1,322	48.9%	0.0250
Male students aged between 18 and 35 years	101	8%	611	22.6%	0.1653
Female non-students aged between 18 and 35 years	153	12%	253	9.3%	0.6047
Male non- students aged between 18 and 35 years	140	11%	173	6.4%	0.8092
Female students aged over 35 years	0	0%	32	1.2%	0.0000
Male students aged over 35 years	1	0%	35	1.3%	0.0286
Female non-students aged over 35 years	439	34%	83	3.1%	5.2892
Male non-students aged below 35 years	381	30%	54	2.0%	7.0556
Missing values	34	3%	143	5.3%	.
Total	1,282	100%	2,706	100.0%	0.4738

Notes: Both samples are categorized based on three socio-demographic dimensions: gender, age and students.

Table A2: Correlation matrix for the five factors of the Vietnamese sample

		Factor 1 Anxiety	Factor 2 Interest in financial issues	Factor 3 Intuitive decisions	Factor 4 Needs for precautionary savings	Factor 5 Free spending
Factor 2	Interest in financial issues	-0.122***	1			
Factor 3	Intuitive decisions	0.158***	-0.027	1		
Factor 4	Needs for precautionary Savings	0.210***	0.103***	-0.053**	1	
Factor 5	Free spending	0.139***	-0.070**	0.488***	-0.057**	1

Notes: n = 2,706. Pearson correlation, sig (two-tailed); ***=correlation is significant at the 0.01 level; **=correlation is significant at the 0.05 level; ***=correlation is significant at the 0.1 level.

Table A3: Correlation matrix for the five factors of the joint sample

		Factor 1 Anxiety	Factor 2 Interest in financial issues	Factor 3 Intuitive decisions	Factor 4 Needs for precautionary savings	Factor 5 Free spending
Factor 2	Interest in financial issues	-0.233***	1			
Factor 3	Intuitive decisions	0.000	-0.013	1		
Factor 4	Needs for precautionary savings	0.175***	0.081***	-0.088***	1	
Factor 5	Free spending	0.097***	-0.103***	0.395***	-0.111***	1

Notes: n = 3,968. Pearson correlation, sig (two-tailed). ***=correlation is significant at the 0.01 level; **=correlation is significant at the 0.05 level; **=correlation is significant at the 0.1 level.

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